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The Enigma Machine: German Use and British Codebreaking

Introduction

Enigma was an electromechanical system of ciphers that the German military and government used from the late 1920s to 1945 to encrypt their communications by radio. Germany felt they were unbeatable. Instead in 1932-45, the Polish and later British codebreakers proved otherwise and their intelligence, dubbed Ultra, influenced the course of the Battle of the Atlantic, North Africa and the Normandy landings.



Figure 1: Military Model Enigma I, in use from 1930

This report describes how Enigma operated, how the German military employed it, its weaknesses and how Poland and Britain outfoxed it.

How the Enigma machine worked

Enigma replaced each typed letter with a different lamp-lit letter, with the substitution varying as a function of the key struck. When the same machine was set up in the same way, typing the ciphertext illuminated the plaintext.

A keystroke was fed into a plugboard, through three rotors into a reflector, back through the rotors and into the plugboard and finally a lamp. Each key press caused the rightmost rotor to advance, causing the substitution alphabet to change (1, 2).

Component	Role	Consequence
Rotors	Wired scramblers that step with each key press; the army and air force picked three from a set of five or more	17,576 starting positions for a three-rotor machine, and the order of rotors was also part of the key
Reflector	Sends the current back through the rotors in pairs	Makes encryption and decryption identical, but means no letter can ever encrypt to itself
Plugboard (Stecker)	Swaps pairs of letters before and after the rotors	Multiplied the number of possible settings by about 150 trillion
Daily key sheet	Listed rotor order, ring settings and plugboard pairs for each day	Every operator on a network used the same settings for the day

In total, the military 3-rotor Enigma could have approximately 1.6 x 10^20 possible settings. The Germans knew that no one could test that many by hand and they were correct. Their weakness was elsewhere.

How Germany used Enigma

The secret of Enigma was the ability to protect radio messages which had to be sent over the air and could be read by anyone. The army, air force and navy each operated their own network using their own key sheets and the machine was used by diplomats, intelligence agencies and the railways (16).

Adoption: A commercial Enigma went on sale in the 1920s. In 1926 it was purchased and converted into a navy vessel; the army would follow suit with its own version in 1928. In 1935, the air force followed the army pattern (18).

Army and air force: The rotors, ring settings and plugboard are installed on the machine from a daily key sheet by operators. In addition, for each message they selected a new message key, to ensure that no two messages on a particular day started from the same rotor positions. These procedures for messages keys were tightened by the army and air force in 1940 (3).

Navy: Enigma was deemed the most important of the navy's systems. It had additional rotors (VI, VII and VIII), and much more involved procedures than the army. Each u-boot had its own key network, and short coded signals of weather and contacts were enciphered before being passed through the machine (5, 4).

The U-boat war: Radio was used for directing U-boats in wolf packs against allied convoys, by Admiral Karl Dönitz. The losses mounted and, despite assurances from German intelligence, he sought increased security. In February 1942 a new model, the M4, was introduced for the navy with a new key network which Bletchley Park named Shark (4).

The weaknesses that enabled codebreaking to be achieved

There were some weaknesses in Enigma's design, but codebreakers could take advantage of each of these.

There will not be any letters that are encrypted to themselves: This is ensured by using a reflector. Such a guessed plaintext text can thus be rejected at any location where a letter would fall on top of another letter (17).

Reciprocal plugboard: In the diagonal board (2), Gordon Welchman later took advantage of this symmetry.

Predictable messages: Fixed phrases were used for opening, for example, a weather report or "nothing to report", and operators repeated the signal on the same topic. These were used by the British to guess plaintext, called cribs (13, 16).

Message-key habits: Until 1940 the army used to send each message key twice at the beginning of a message, resulting in patterns which the Poles exploited (3).

Captured material: In the absence of dedicated mathematics, the British received code books, key sheets and U-boat and ship code machines.

German faith was also a liability. The designers thought the key space made the machine safe and therefore were slow to suspect that their messages were being read.

The Polish breakthrough

Britain was not a tabula rasa. In December 1932, Polish mathematicians, almost seven years before the outbreak of war (9), successfully broke the military Enigma.

Instead of guesswork, Marian Rejewski, with Jerzy Różycki and Henryk Zygalski of the Polish Cipher Bureau used permutation mathematics. By the end of 1932 Rejewski was able to decipher the wiring of the military rotors, having been provided with it by French intelligence in the form of manuals and key sheets (6). The team then fabricated replicas of the machine, and developed a method for recalling the daily program, which involved a catalogue of cycle patterns, followed by the electromechanical Bomba, designed in October 1938 (7, 9).

Germany continued to increase the complexity, such as additional rotors and additional plugboard cables, and by 1939, the Polish methods were becoming ineffective. When invasion was imminent the Cipher Bureau informed the French and British of its actions and handed over Enigma replicas and its methods (8, 9). Rejewski and his colleagues then fled to Britain (7).

The Polish work provided Bletchley Park with a starting point and demonstrated to the Allies that mathematicians were the right people to recruit.

How Bletchley Park broke Enigma

During the war the Government Code and Cypher School relocated to Bletchley Park, where Britain's codebreakers operated. Alan Turing, who led the naval effort on Enigma in Hut 8 (11), arrived on 4 September 1939, while Gordon Welchman was head of the army and air force work in Hut 6. The key to their success lay in four areas.

Cribs: The likely plaintext was made up of daily routines like weather forecasts and "Nothing to report." A whole team at Bletchley was involved in locating cribs and lining them up with the intercepted ciphertext (12, 13).

The Bombe: Turing realized that the test for a crib to be capable of being run through all the settings of the rotors could be turned into a machine. The machine automatically found settings that were not in conflict with the crib and delivered the survivors to a human for verification (10). In March 1940, the first Bombe was put into operation.

The diagonal board: The reciprocity of the plugboard was used by Welchman to eliminate spurious results, and to make the Bombe much faster. At the end of the war 211 Bombes were in use (10).

Captured material: The navy's machines were equipped with additional rotors and more stringent operating guidelines, making cribs hard to come by. The boarding party of HMS Bulldog were able to board U-110 on 9 May 1941 and remove Enigma and code books. Bletchley had already got some excellent material from weather ships, and the captured books had enabled him to read the U-boat traffic regularly (14, 4).

The four-rotor M4 and Shark key arrived in February 1942, and Bletchley lost the U-boat traffic for a period of some 10 months. It broke back when more material from the sunken U-boat was recovered, at the cost of two lives (4). The British made use of a statistical technique called Banburismus that reduced the number of rotor orders that the Bombe had to test, which was developed by previous Polish research (9).

Ultra and its impact

Enigma traffic deciphered and intelligence was known as Ultra, which was one of the most closely guarded secrets of the Allies. The Allies at the height of their powers could read about 3000 German messages daily (20).

Battle of the Atlantic: The Admiralty used to be able to decipher the U-boat signals and direct convoys away from wolf packs; and later to locate the boats (15). Over the war, over 700 U-boats were sunk with more than half of those after 1943 (19).

Other fronts: In 1943 and 1944 (16), Bletchley decrypts provided information about German weapons research and about defenses on the Atlantic Wall.

Secrecy: The Allies would sometimes mask their knowledge of something to preserve Ultra. U-110 then disappeared after that point, and the Admiralty made it appear that it was merely sunk.

Historians have divided the opinion as to how much Ultra shortened the war. While the general rule is two years, it is difficult to substantiate, and industrial strength, air power and the convoy system, in turn, played a big part (22).

Conclusion

Enigma was feared and trusted by Germany because it has a lot of settings but the machine itself and the way it was used gives out information. Poland got in mathematical ways and Britain made that into a working system of cribs, Bombes, captured material and statistical methods. The teaching point for cryptography is that a large key space is not sufficient defense for a system whose structure and everyday use provide the attacker with some "cracking tools".

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